PSOC™ Edge E84 training manual: Extended Boot

About this document

Scope and purpose

This document is a training manual for the hands-on exercise with an Extended Boot on PSOC™ Edge using ModusToolbox™.

Intended audience

The document is intended for design engineers, technicians, and developers of electronic systems.

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1 Introduction

This manual provides specific instructions to create, configure, build, and run the code examples on the PSOC™ Edge E84. In this lab session, you will learn:

- How to replace the Extended Boot on the device with a new version
- Extended Boot capability to copy an image from the external flash to SRAM and launch it from SRAM

2 Required development tools

- ModusToolbox™ software v3.6 or later
  - Recommended installation via ModusToolbox™ Setup tool
- Eclipse IDE for ModusToolbox™ v2025.4.0 or later
  - Recommended installation via ModusToolbox™ Setup tool
- Edge Protect Security Suite v1.6 or later.
  - Installed by ModusToolbox™ Setup tool as a dependency to ModusToolbox™ v3.6
- ModusToolbox™ Programming Tools v1.5.0 or later
  - Installed by ModusToolbox™ Setup tool as a dependency to ModusToolbox™ v3.6
- Board support package (BSP):
  - KIT_PSE84_EVAL_EPC2: v1.0.0 or later (available with ModusToolbox™ v3.6)
- Terminal emulator
  - Instructions in this document use Tera Term
- PSOC™ Edge E84 Evaluation Kit (KIT_PSE84_EVAL)
  - Ensure device boots from external memory:
    - BOOT SW/SW6 in 'ON/HIGH' position
  - Disable alternate serial interface configuration:
    - J20/J21 in Tristate/Not-Connected (NC) position

3 Prerequisites

Install all required software and hardware listed in the Required development tools section.

Note that this class will not cover the basics of ModusToolbox™ and PSOC™ Edge.

- For an introduction to PSOC™, including a getting started guide to ModusToolbox™, visit: https://www.infineon.com/product-information/psocdeveloper
- For PSOC™ Edge trainings, from beginner tutorials to advanced trainings, visit: PSOC™ Edge E84 Training Collection

4 Replacing Extended Boot

4.1 Pre-requisite

To replace the Extended Boot, you should have already taken ownership of the device via the provisioning process. **Training Manual for getting started with PSOC™ Edge E84 Security using ModusToolbox™** provides detailed step-by-step instructions. If you have not already followed those steps, please complete them before getting further into this lab session.

4.2 Objective

The objective of this lab session is to understand how to replace the Extended Boot on PSOC™ Edge devices. You can replace the Extended Boot on the device with the version that is shipped with Edge Protect Tools.

4.3 Detailed solution

1. Follow the step-by-step instructions provided in the **Training Manual for getting started with PSOC™ Edge E84 Security using Modus Toolbox™** guide to obtain the device ownership. If you have already completed the steps to obtain ownership, you need not re-obtain the ownership.
2. Run the command below to observe the version of the Extended Boot.

```
edgeprotecttools -t pse84 device-info
```

3. Open the policy file *policy_oem_provisioning.json* located in *<application-directory>/policy/* directory.
4. Update the *extended_boot_image* policy with an absolute path to the Extended Boot image. For example, see the “value” field in the policy below, which points to the new version of the Extended Boot image.

```
"extended_boot_image": {
    "description": "Image of Extended Boot in CBOR format, signed by IFX",

"value":"C:/WS/PSOC_Edge_Basic_Secure_App/packets/apps/prov_oem/extended-
boot- serial.cbor"
}
```

5. Now, provision the device using the following command and observe the log message that indicates provisioning completed successfully.

```
edgeprotecttools -t pse8xs2 provision-device -p
policy/policy_oem_provisioning.json --key keys/oem_private_key_0.pem --ifx-
oem-cert packets/apps/prov_oem/oem_cert.bin
```

```
: O : INFO  : ** Detected Device: PSE846GPS2DBZC4A
: O : INFO  : ** SROM Boot version: 2.0.0.6022
: O : INFO  : ** RRAM Boot version: 2.0.0.7127
: O : INFO  : ** SE RT Services Base version: 1.0.0.2361
: O : INFO  : ** SE RT Services version: 0.0.0.0
: O : INFO  : ** Extended Boot version: 1.1.0.1700
: O : INFO  : ** Boot Status : CYBOOT_SUCCESS
: O : INFO  : ** Life Cycle  : DEVELOPMENT
: O : INFO  : Info : [cat1d.sys] Examination succeed
: O : INFO  : Info : gdb port disabled
: O : INFO  :     TargetName         Type       Endian TapName            State
: O : INFO  : --  ------------------ ---------- ------ ------------------ ------------
: O : INFO  :  0* cat1d.sys          mem_ap     little cat1d.cpu          halted
: O : INFO  : Info : Listening on port 6668 for tcl connections
: E : INFO  : Chip lifecycle stage: DEVELOPMENT
: E : INFO  : ROM Boot is ready for DLM package upload
: E : INFO  : Load application 'C:\workspace\Security_Intro\Basic_secure_app\packets\apps\prov_oem\cyapp_prov_oem_dlm.he
x'
: E : INFO  : Validation of the DLM package success
: E : INFO  : Start DLM package execution
: E : INFO  : Application execution completed with status code: (0xf2a00001) - CYAPP_SUCCESS: The provisioning applicati
on completed successfully
: E : INFO  : Chip lifecycle stage: DEVELOPMENT
: E : INFO  : *****************************************
: E : INFO  :            PROVISIONING PASSED
: E : INFO  : *****************************************
```

Figure 1 Terminal output for provisioning

6. On the successful completion of the provisioning, the Extended Boot is replaced. The updated version of the Extended Boot will be displayed on the console, as shown in Figure 2. Compare this version with the one shown before the replacement to confirm the update.

```
: INFO  : ** Silicon: 0xED94, Family: 0x115, Rev.: 0x21 (B0)
: INFO  : ** Detected Device: PSE846GPS2DBZC4A
: INFO  : ** SROM Boot version: 2.0.0.6022
: INFO  : ** RRAM Boot version: 2.0.0.7127
: INFO  : ** SE RT Services Base version: 1.0.0.2361
: INFO  : ** SE RT Services version: 0.0.0.0
: INFO  : ** Extended Boot version: 1.1.0.1700
: INFO  : ** Boot Status : CYBOOT_SUCCESS
: INFO  : ** Life Cycle  : DEVELOPMENT
```

Figure 2 Extended Boot version

Note: Replacement of Extended Boot is not necessarily an update or downgrade of Extended Boot. You can replace the image on the device with the same version.

4.3.1 Conclusion

The Extended Boot on the device has been replaced with the latest version provided in the Edge Protect Tools.

5 SRAM loading of the user application using the Extended Boot

5.1 Prerequisite

You should have completed the ownership transfer via the provisioning process in the previous lab, **Training manual for getting started with PSOC™ Edge E84 Security using ModusToolbox™**.

5.2 Objective

By default, the 'Basic Secure Application' code example is configured to program and run from external flash. All three projects (proj_cm33_s, proj_cm33_ns, and proj_cm55) in this code example are built, programmed, and executed in-place from external flash. Upon reset, the Extended Boot searches for a valid image (in MCUboot format) in the predefined location and launches it. For the Extended Boot to search the image in external flash, the boot source must be set to external flash.

In this lab session, let us learn how to load a project from an external flash to SRAM and launch it in SRAM. Before that, let us first understand the fundamental design concepts for this use case.

5.3 Description

The 'Basic Secure Application' code example is built to boot from the external flash (XIP).

To select the boot source in external flash, the BOOT Pin and Boot Policy must be set accordingly.

1. Default policy defines oem_alt_app_address as 0x7010_0000, which is in external flash. This is the default start address of the application if external flash is selected as a boot source.
2. Default policy also enables oem_alt_boot, which indicates alternate boot option is enabled.
3. To select external flash as the boot source, BOOT.1 Pin (BOOT SW) (P17.6) must be in 'ON/HIGH' position.

Note: This is a DIY (Do it Yourself) lab exercise. Look into the detailed solution below, only if you need guidance.

5.4 Detailed solution

5.4.1 Creating the project

This hands-on session uses the 'Basic Secure Application' code example. This is the same code example that is used in the "*Introduction to PSOC™ Edge Security*" training. See the "*Training Manual for getting started with PSOC™ Edge E84 Security using ModusToolbox™*" for more details.

To create a new Basic Secure code example, follow the detailed instructions provided in the **Introduction to PSOC™ Edge Security** lab session.

Note: The code example is configured to use the GCC_ARM compiler by default. We use the same compiler in this lab session.

5.4.2 Configure proj_cm33_s to run from SRAM

In the default configuration, proj_cm33_s of Basic Secure Application is built to run from an external flash. Modify the default memory map and linker to run it from SRAM.

1. Select 'Basic Secure App' in the **Project Explorer** and open the **Device configurator** from the Quick Panel.

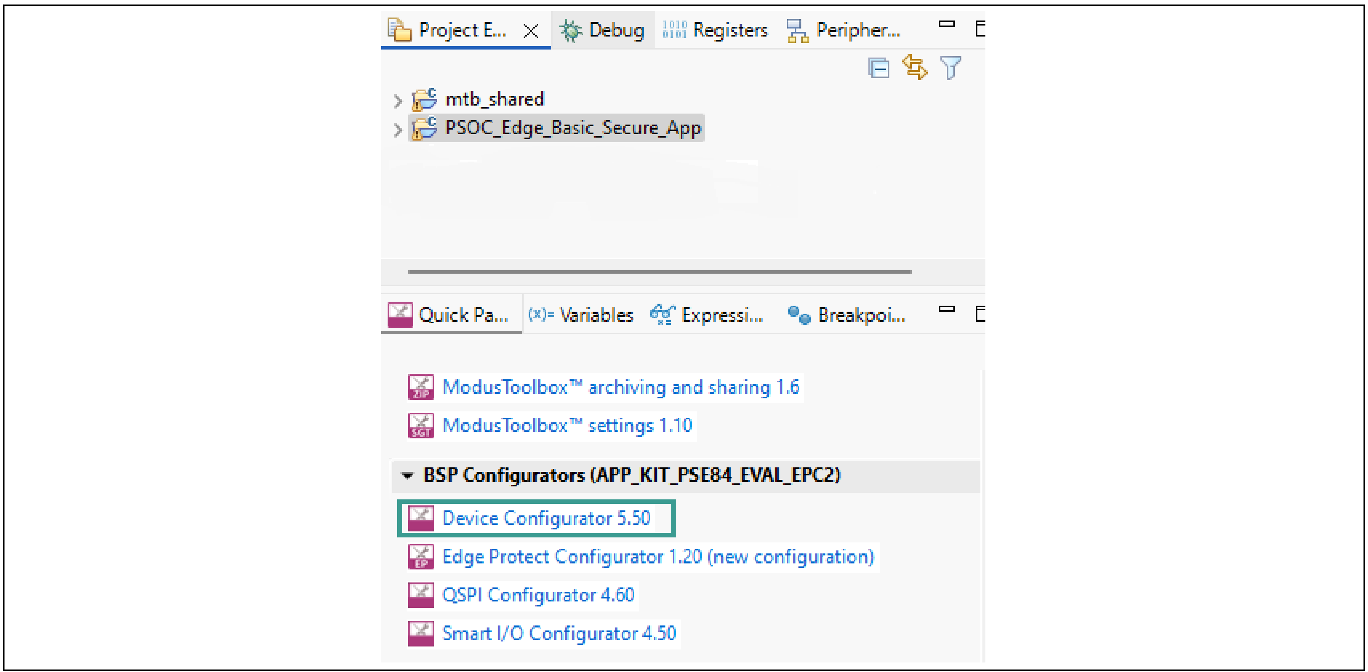

Figure 4 Open Device Configurator

2. Under the **Memory** tab, add a new region in SRAM for loading the CM33 S project.

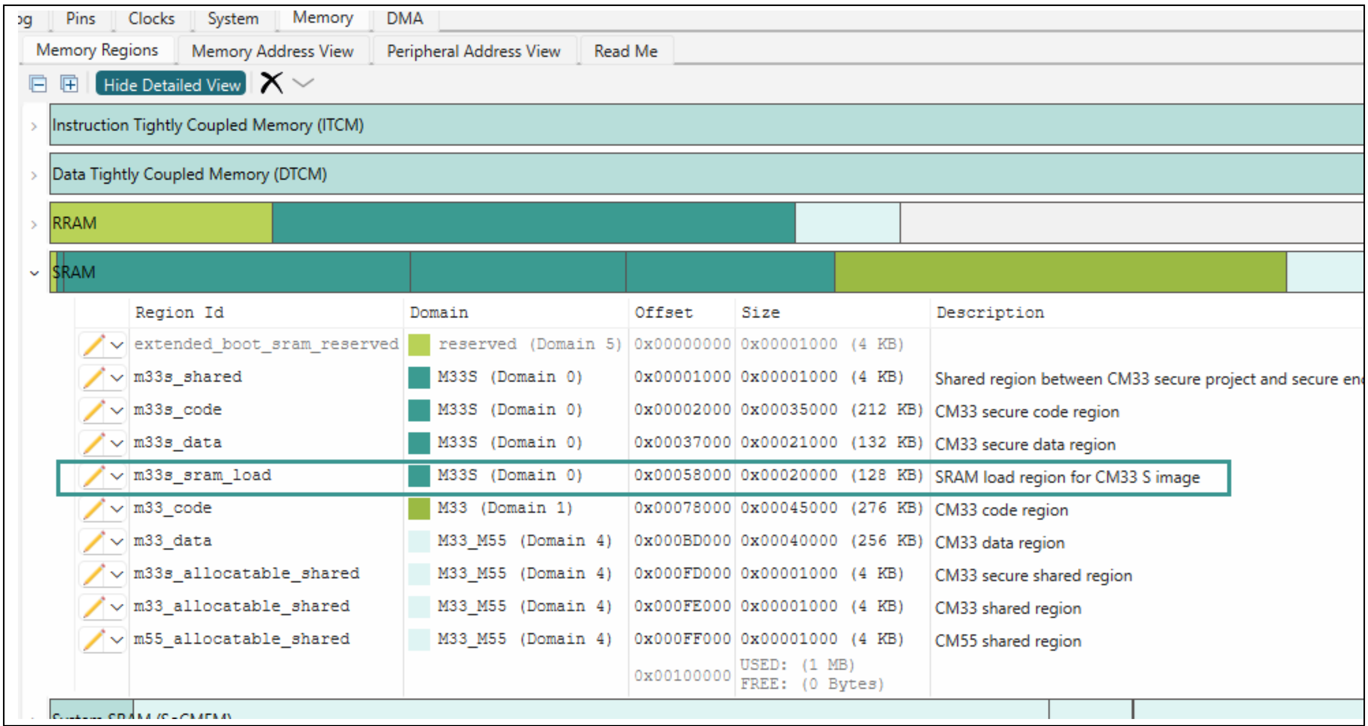

Figure 3 Create a new region for SRAM loading

SRAM loading of the user application using the Extended Boot

Note: You may need to adjust the existing regions in the default memory map to make space.

3. Save the configuration in the **Device Configurator** and close it.
4. In the Basic Secure Application, open the cm33s linker file in editor *<pathto>PSOC_Edge_Basic_Secure_App/bsps/TARGET_APP_KIT_PSE84_EVAL_EPC2/COMPONENT_CM33/TOOLCHAIN_GCC_ARM/pse84_s_cm33.ld* and make the following change in linker file.
5. Find and replace all **m33s_nvm_sel_S** with **m33s_sram_load_sel_S**, and save your changes.

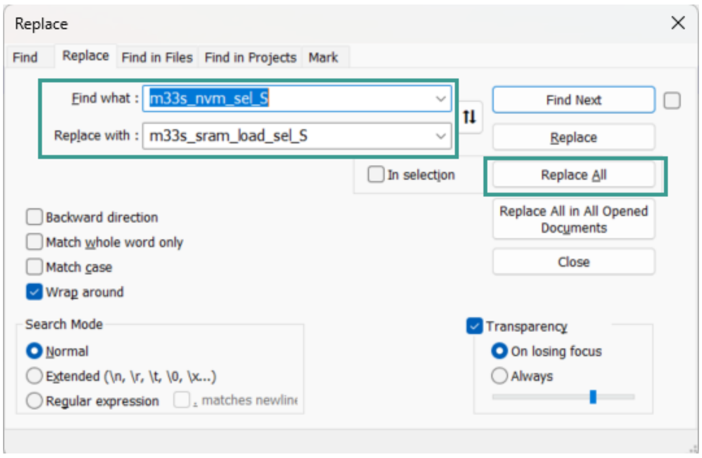

Figure 4 Edit CM33 S linker

6. Open the file *<pathto>PSOC_Edge_Basic_Secure_App/configs/boot_with_extended_boot.json* and add the following additional "load-address" input for the sign command and save.

```
[
    "schema-version": 1.0,
    "content":
    [
        {
            "name": "metadata_proj_cm33_s",
            "description": "add MCUboot metadata to the hex",
            "enabled" : true,
            "commands" :
            [
                {
                    "command" : "sign",
                    "inputs" :
                    [
                        {
                            "file" : "../build/project_hex/proj_cm33_s.hex",
                            "header-size": "0x400",
                            "fill-value" : "0xff",
                            "slot-size" : "0x240000",
                            "hex-address" : "{{CYMEM_CM33_0_S_m33s_nvm_S_START}}",
                            "load-address" : "{{CYMEM_CM33_0_S_m33s_sram_load_S_START}}"
                        }
                    ],
                    "outputs":
                    [
                        {
                            "file" : "../build/project_hex/proj_cm33_s_signed.hex",
                            "format": "ihex"
                        }
                    ],
                    "extra_config":
                    [
                        {
                            "project": "proj_cm33_s",
                            "debug_config_name" : "proj_cm33_s",
                            "default":false,
                            "build_dependency" : "project"
                        }
                    ]
```

Figure 5 Edit the combiner signer JSON file

7. Build and program the Basic Secure Application in the ModusToolbox™.

Note: *The Instructions to build and program are the same as the previous lab in the previous session – "Introduction to PSOC™ Edge Security". See the "Training Manual for getting started with PSOC™ Edge E84 Security using ModusToolbox™" for more details.*

8. Ensure that the BOOT.1 pin (SW6) is in HIGH/ON position. This is required for the Extended Boot to determine the external flash as the boot source.

5.4.3 Observing the results

1. Observe the console logs for successful booting of the Basic Secure Application.
2. Observe that the red LED is blinking on the device.

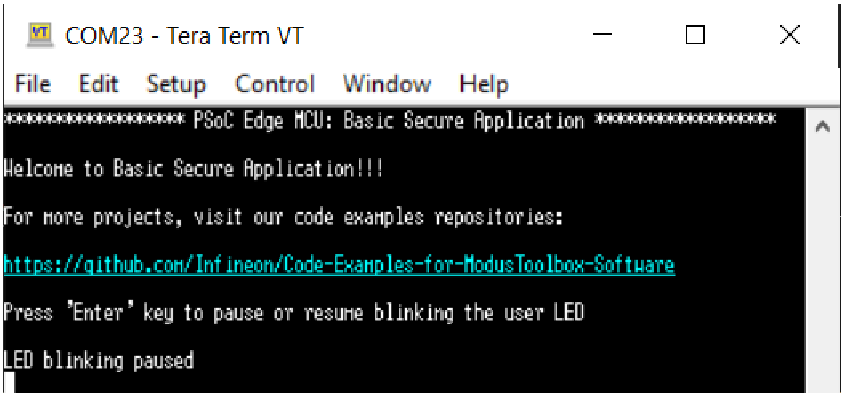

Figure 6 Terminal output

5.5 Conclusion

The secure project (proj_cm33_s) in the Basic Secure Application is programmed to the external flash. The Extended Boot copies it to the SRAM and launches it from there.

Revision history

Document revision	Date	Description of changes
**	2025-12-10	Initial release

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